

**VCH Reproducible Research Framework
Design and Justification**

Table 1 – Final Reproducible Research Framework for VCH and Windesheim

Step	What happens in practice at VCH	Minimum required outputs	Who is responsible	When it is checked	How success is tested
1 Project workspace setup before research starts	The project is created inside a fixed digital structure before any data is collected or analysed	Approved folder structure, decision log file, source log file, one page project summary file, naming rules applied	Student project lead sets up, supervisor approves	Before any research begins	Research cannot start unless all required folders and files exist and follow the agreed structure
2 Research plan and source definition	The team clearly defines what data will be used, where it comes from, and why it is relevant	Short methods summary, list of data types, list of key source categories, explanation of how each will be stored, active source log with purpose field completed	Student project lead drafts, supervisor approves	Before fieldwork or deep desk research begins	All intended data types and source categories are written down with purpose and storage location

3 Data collection and structured analysis during execution	Data is collected and analysed at the same time. Students compare findings, identify patterns, combine desk and field insights, and write down their reasoning as it develops	Updated source log entries, decision log entries for every major interpretation shift, stored interview notes or datasets, written explanation of how evidence supports emerging conclusions, stored analysis outputs such as comparison tables or summaries	Student project lead maintains weekly, supervisor reviews during progress meetings	Ongoing during project execution	At least two developing conclusions can be directly linked to stored data and written reasoning notes. All major changes in interpretation include written explanation
4 Version tracking and development stability	As the report develops, earlier versions are kept so the path from draft to final report is visible	Date stamped report versions, short change summary per version, preserved earlier drafts, exported copies from external tools so nothing disappears	Student project lead maintains, supervisor checks at midpoint	Midpoint review before finalisation	No file is overwritten. A reviewer can follow the development path from early draft to final version step by step

5 Reconstructability verification gate	Before submission, someone tests whether the reasoning can be reconstructed without asking the team	Completed reconstruction checklist, written test showing how two key conclusions were rebuilt from data and reasoning	Verification reviewer performs, supervisor authorises submission	Before final submission	A person not involved in the project can explain what data was used, what reasoning steps were taken, and how at least two final conclusions were formed
6 Dual submission and structured handover	The project is submitted as both a report and a full documentation package. It is then archived for future cohorts	Final report, decision log, source log, one page archive summary, stored datasets where allowed, handover brief listing open questions and continuation suggestions, confirmed archive location	Supervisor confirms completeness, lab coordinator oversees archive	At submission and at start of next cohort	Submission is incomplete without a documentation package. Next cohort can reuse at least one prior conclusion using only archived materials

Table 2 – Theoretical and Research Justification

Table 2

Theoretical and Empirical Justification of Each Step

Step	What problem from field research it fixes	Theoretical foundation explained clearly	Why that theory matters here	External institutional practice it mirrors	Link to the research questions
1 Workspace setup before research	Projects stored files inconsistently or lacked clear structure, which made reuse and verification difficult	Work System Theory explains that behaviour follows structure. If no structure is provided, people improvise. Governance theory explains that documentation must be built into the system rather than left to personal motivation.	Without early structure, documentation becomes optional and inconsistent. Creating structure first reduces variation and prevents future loss.	Mature university research environments require structured repositories and defined metadata before research begins.	Subquestion 1 identifies structural weaknesses. Subquestion 2 benchmarks stronger institutional systems. Subquestion 3 translates those standards into operational design.
2 Research plan and source definition	Links between sources and conclusions were sometimes unclear, and reasoning was reconstructed only at the reporting stage	Provenance theory states that every claim must be traceable back to its origin. Lifecycle thinking shows that traceability must start at the beginning, not at the end.	If sources are not defined early, later claims cannot be audited or verified. This step prevents invisible reasoning chains.	Research data management standards require early documentation of what data will be used and how it will be handled.	Subquestion 1 identifies missing traceability. Subquestion 3 turns traceability into a required structure.

3 Data collection and structured analysis during execution	Decision shifts and analytical reasoning were often not recorded, making it difficult to reconstruct how conclusions were formed	Reproducibility research explains that reasoning must be visible, not only final results. Knowledge externalisation theory explains that ideas must be written down to become shared knowledge. Research quality standards require explicit reasoning and pattern identification.	Analysis is the core of research. If interpretation is not documented while it happens, future readers cannot understand how insights emerged.	Structured interview protocols and reproducibility initiatives emphasise documenting decisions and reasoning during execution.	Subquestion 1 identifies reasoning gaps. Subquestion 3 designs embedded documentation mechanisms. Subquestion 4 requires measurable validation of reasoning visibility.
4 Version tracking and development stability	Multiple report versions existed without clear progression, weakening transparency of how conclusions evolved	Design Science theory explains that knowledge develops through iteration. To evaluate development properly, earlier versions must be preserved.	If earlier drafts are lost, the evolution of reasoning disappears. Version tracking makes analytical development visible and reviewable.	Institutional repositories require stable file naming and preserved versions to maintain long term accessibility.	Subquestion 3 translates reproducibility principles into concrete structural mechanisms.
5 Reconstructability verification gate	A readable report did not always mean the reasoning behind it was verifiable	Provenance logic defines reconstructability as the ability to independently rebuild reasoning from stored information. Design Science requires evaluation against explicit criteria before claiming success.	Without testing reconstructability, documentation may appear complete but still fail under scrutiny. This step makes quality measurable.	Reproducibility initiatives in universities include review routines that test whether reasoning can be independently understood.	Subquestion 4 focuses on execution and evaluation. This step operationalises measurable validation.

6 Dual submission and structured handover

Continuity between cohorts was weak, and reuse often depended on informal explanation rather than structured documentation

Communities of Practice theory explains that knowledge survives when embedded in shared systems and routines. FAIR principles emphasise reuse as a measurable condition of quality.

If projects are submitted as reports only, knowledge remains isolated. Structured handover transforms isolated work into cumulative institutional knowledge.

University data platforms and structured archives enable long term reuse through metadata and organised storage.

Subquestion 5 focuses on long term continuity and institutional sustainability.